

Lecture Notes: Mathematical Physics Equations and Special Functions

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Week 5-2

3.2 Initial Value Problems for Higher Dimensional Wave Equations

Key Ideas

- The approach starts by considering the 3D wave equation and then applying dimensional reduction to handle the 2D case.
- The main difficulty lies in the 3D wave equation, while the 2D case is relatively easier.
- The nature of solutions differs significantly between even and odd spatial dimensions for wave equations.
- Despite deriving 2D results from the 3D case, their solution properties remain fundamentally different.

Physical Analogy: Gravity vs. Wave Propagation

- Gravity is governed by Poisson's equation, which does not evolve over time.
- A thought experiment: Suppose ϕ satisfies the 3D wave equation instead.
- General Relativity suggests that gravity behaves like waves, though in a much more complex manner.
- This leads to an intuitive approach: **body thinking**.
- If gravity were governed by a wave equation, it would exhibit oscillatory behavior.
- This would mean that gravitational strength at a point would fluctuate over time, akin to experiencing a roller-coaster effect.

Simplest 3D Wave: Spherical Symmetry

- The simplest 3D wave solution is a **spherically symmetric wave**.
- Spherically symmetric waves simplify the problem by **reducing dependence on angular coordinates**.
- This allows for a transformation of the problem into an effectively 1D equation with a **radial coordinate**.

Body Thinking: A Learning Strategy

- Body thinking involves associating abstract concepts with physical sensations.
- Example: A chemist mentally links chemical reactions to body movements, responding physically to conceptual changes.
- Applying this idea: Imagine being surrounded by a fluctuating gravitational field.
- In this model, gravity varies dynamically like a wave:
 - You feel gravitational strength increasing and decreasing.
 - The sensation is similar to riding a roller coaster while sitting still.

Approach to the 3D Wave Equation

- Direct solution methods such as separation of variables are not feasible without boundary conditions.
- The only viable approach is to transform the 3D problem into a simpler 1D form.
- However, reducing 3D to 1D is challenging due to the greater degrees of freedom in 3D.
- To solve the 3D wave equation, we begin by identifying the simplest case: **spherical symmetry**.
- The problem-solving approach resembles detective work:
 - Collect key clues.
 - Formulate bold hypotheses.
 - Attempt to construct a viable solution.
- The first clue: consider the simplest wave solution, a spherically symmetric wave.
- Using the **method of dissecting complexity** (similar to *Pao Ding's Butchering the Ox* analogy), we rewrite the wave equation in spherical coordinates.
- This transformation simplifies the problem and helps in deriving an explicit solution.

In the previous section, we discussed the initial value problem of the **one-dimensional** wave equation and obtained **D'Alembert's formula**. For the **three-dimensional** wave equation, the solution can be expressed in a **spherical mean form**, which is commonly referred to as **Kirchhoff's formula**.

3.2.1 Kirchhoff's Formula for the Three-Dimensional Wave Equation

Now, let's consider the initial value problem of the three-dimensional wave equation

$$u_{tt} = a^2(u_{xx} + u_{yy} + u_{zz})(-\infty < x, y, z < +\infty, t > 0), \quad (1)$$

$$u(x, y, z, 0) = \varphi(x, y, z), \quad u_t(x, y, z, 0) = \psi(x, y, z), \quad (2)$$

where $\varphi(x, y, z)$ and $\psi(x, y, z)$ are known functions.

How to invent the method of spherical mean

Clue 1:

To explain how the spherical means method is conceived, let's first look at the **first clue**: the spherically symmetric solution of the three-dimensional wave equation.

Using spherical coordinates:

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases}$$

The **wave equation in spherical coordinates** (by chain rules) is expressed as:

$$\Delta u = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} = \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2}$$

Since u is **independent of θ and ϕ** (due to the spherical symmetry)

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2}$$

- We currently have the 1D wave equation describing string vibrations.
- The only available approach is the **traveling wave method**, which provides solutions in the form of moving waves.
- Our goal: Transform this equation into the **1D wave equation** to simplify analysis.

Recall a useful relation

$$\partial_r(r^2 \partial_r u) = r \partial_r^2(ru) \quad (3)$$

Then we arrive at

$$\frac{\partial^2(ru)}{\partial r^2} = \frac{1}{a^2} \frac{\partial^2(ru)}{\partial t^2}$$

The general solution gives:

$$ru = f(r - at) + g(r + at)$$

That is,

$$u = \frac{f(r - at)}{r} + \frac{g(r + at)}{r}.$$

This is the **spherically symmetric** solution of the three-dimensional wave equation.

Clue 2:

Using the D'Alembert's formula to **guess** the 3D formula,

- The goal is to rewrite the formula to unify two types of averages:
 - **Arithmetic Mean**
 - **Integral Mean**
- Mathematicians prefer **symmetry** and **consistency**, leading to the search for a **unified form** ← **Aesthetic Criterion**.
- The **arithmetic mean** can be expressed as an **integral mean** by integration followed by differentiation.

$$\begin{aligned}
u(x, t) &= \frac{1}{2} (\varphi(x - at) + \varphi(x + at)) + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\alpha) d\alpha \\
&= \frac{1}{2a} \partial_t \left(\int_{x-at}^{x+at} \varphi(\alpha) d\alpha \right) + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\alpha) d\alpha \\
&= \frac{\partial}{\partial t} \left(\frac{t}{2at} \int_{x-at}^{x+at} \varphi(\alpha) d\alpha \right) + \frac{t}{2at} \int_{x-at}^{x+at} \psi(\alpha) d\alpha
\end{aligned}$$

- The rewritten formula contains:
 - The integral mean of the initial displacement.
 - The integral mean of the initial velocity, scaled by time.
- Since the 1D and 3D cases are special cases of an n -dimensional wave equation, there must be an underlying relationship.
- Extending the 1D case to 3D leads to a natural question:
 - What type of averaging should be used in 3D?
 - Possibilities: **Spherical volume average** vs. **Spherical surface average**.
- Experimental verification (trial and errors) shows that **spherical surface averaging** is the appropriate choice for solving the 3D wave equation.

Analogy and **guess** (see Fig. 1):

$$u(M, t) = \frac{\partial}{\partial t} \left(\frac{t}{4\pi(at)^2} \int_{S_{at}^M} \varphi(\xi, \eta, \zeta) dS \right) + \frac{t}{4\pi(at)^2} \int_{S_{at}^M} \psi(\xi, \eta, \zeta) dS \leftarrow \boxed{\text{Aesthetic Criteria}} \quad (4)$$

This may be the solution to the three-dimensional wave equation, and

- It suggests that the solution **at a given point** can be expressed using the **spherical average around that point**.

- Conclusion:
 - **Spherical symmetry** is a crucial property in solving the 3D wave equation.
 - **Spherical surface averaging around every point** plays a fundamental role in the solution process.

About (4)

- From the conjectured formula, in order to determine $u(t_0, x_0)$, we need to know the data on the sphere at any arbitrary time t_1 . However, in reality, we only need the spherical average data at any given time t_1 . This means that if we know the spherical average at any time t_1 , it is sufficient to determine $u(t_0, x_0)$.
- However, the direct result of the spherical average data is the spherical average solution $\bar{u}$

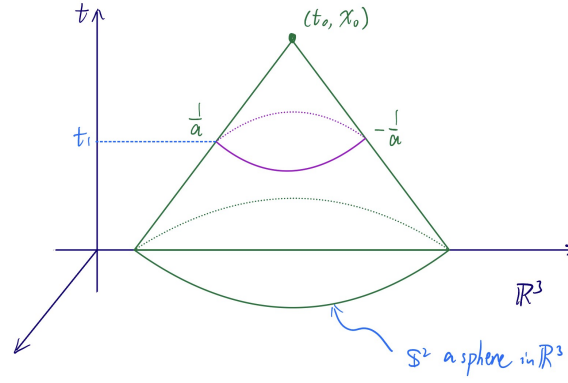
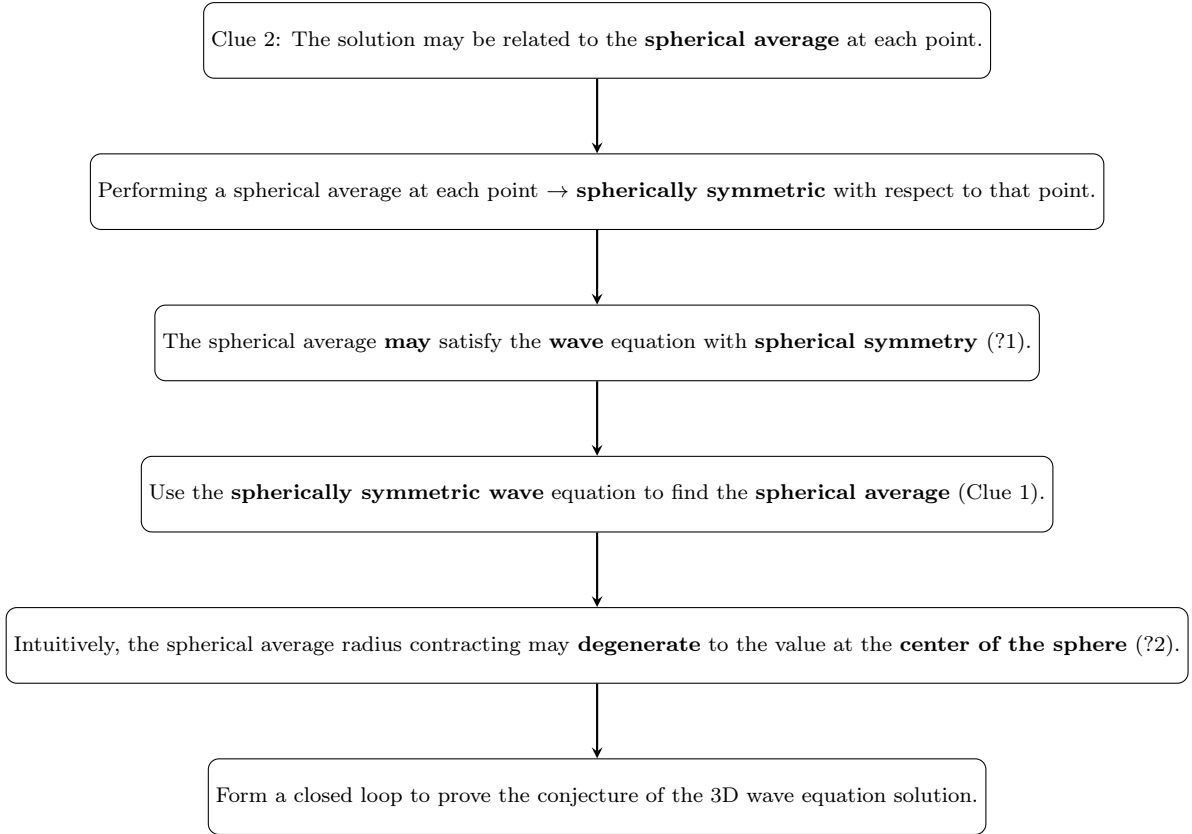


Figure 1: 3D wave analogy

instead of u . Therefore, we pose the following questions:

1. What is the relationship between u and $\bar{u}$?
 - The information of u and $\bar{u}$ should be equivalent.
 - * If u is known, $\bar{u}$ can be determined.
 - * Conversely, knowing $\bar{u}$, intuition suggests that as the radius approaches zero, $\bar{u} \rightarrow u$.
 - * This follows from the fact that as the concentric spheres around point M shrink to a single point, the average over the sphere reduces to the value at M .
 - This equivalence allows us to study u via $\bar{u}$.
 - * u is difficult to determine directly.
 - * However, $\bar{u}$ depends only on a single spatial variable r , making it potentially solvable using a one-dimensional wave equation.
 - From the conjectured solution:
 - * u does not distinguish specific initial values but only their spherical averages.
 - * This indicates that redistributing the values on the surface of a sphere such that every point takes the spherical average results in the value at its center remaining unchanged. That is, as long as their spherical averages are the same, the center cannot distinguish which case is which.
 - * This transformation leads to a spherically symmetric problem, which may connect to **Clue 1**.
 2. What equation does $\bar{u}$ satisfy?
 - The information of u and $\bar{u}$ is equivalent (a conjecture that needs to be proven). Since the information of u is equal to the information contained in the wave equation, it follows that the information of $\bar{u}$ should also be equal to the information of the wave equation. Therefore, the wave equation should be able to derive the equation for $\bar{u}$.
- The above considerations are purely based on the conjectures. Based on this idea, we can further refine and develop a mathematical method.

According to these clues, we conceive the ideas:



where (?1) and (?2) are conjectures that we have to prove. Next, we first prove these two questions.

Key Questions and Solutions

How does the spherical average relate to the original solution? (about (?2))

- Consider a sphere of radius r centered at any given point.
- Compute the spherical average over this sphere.
- As $r \rightarrow 0$, the spherical average approaches the value of the function at the center.
- This suggests that the spherical averaging method can be used to approximate the solution at any point.

Does the spherical average satisfy a wave equation? (about (?1))

- Two possible ways to check:
 1. **Assume the solution exists:** If the solution to the wave equation exists everywhere, we can explicitly compute its spherical average.
 2. **Use the wave equation itself:** The wave equation contains all necessary information about the solution, making it a valid tool to derive properties of the spherical average.

Wave Eqs $\xrightarrow{\text{directly derive}}$ Eq. of Spherical Average (spherical symmetric wave eq.)

- Since the wave equation is **equivalent to the information** contained in the solution, if properly **transformed**, it should **hold for the spherical average** as well (like the energy method).

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Original Wave Equation $\Leftrightarrow u \overset{?}{\Leftrightarrow} \bar{u} \Leftrightarrow$ Spherical Mean Equation (Spherically Symmetric Wave Equation)?

First, fix any point $M = (x, y, z)$, S_r^M represents the sphere with center M and radius r . Using spherical coordinates, a point on the sphere is given by:

$$P \equiv (\xi, \eta, \zeta) = (x + r \sin \theta \cos \phi, y + r \sin \theta \sin \phi, z + r \cos \theta).$$

Let $\omega = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ represent the unit outward normal vector to the sphere S_r^M , then a point on the sphere S_r^M can be simply written as $M + r\omega$. At the same time, ω can also be considered as a point on the unit sphere. Therefore, we also denote the surface element on the sphere as:

$$dS_r^M = r^2 \sin \theta d\theta d\phi \quad \text{and} \quad d\omega = \sin \theta d\theta d\phi.$$

Note the normalization relation ($d\omega$ is independent of r and it leads to some convenience).

$$dS_r^M = r^2 d\omega.$$

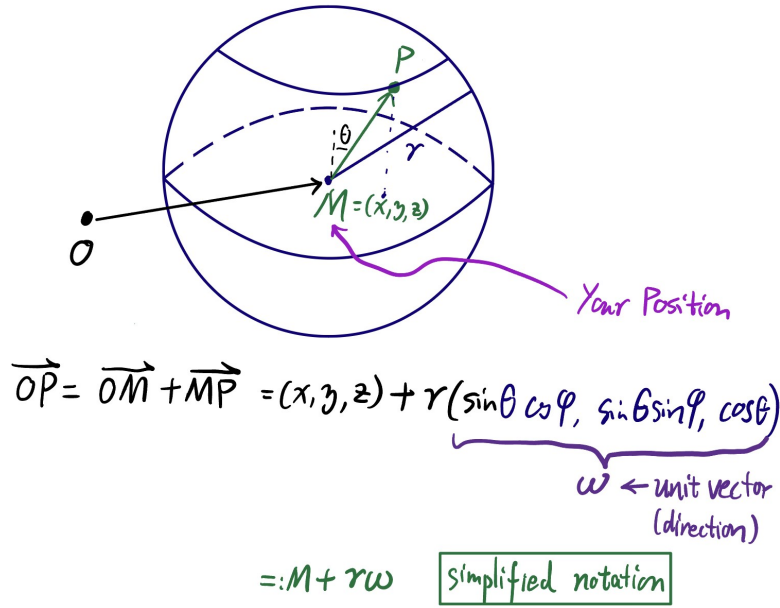


Figure 2: Notation on Sphere

Now introduce the **spherical average** of u (note $dS_r^M = r^2 d\omega$):

$$\bar{u}(r, t) \equiv \frac{1}{4\pi r^2} \iint_{S_r^M} u(P, t) dS_r^M = \frac{1}{4\pi} \iint_{S_1^M} u(M + r\omega, t) d\omega.$$

(Proof of (?2)) Taking the limit as $r \rightarrow 0$ on both sides of the above equation, we get:

$$\lim_{r \rightarrow 0} \bar{u}(r, t) = \frac{1}{4\pi} \iint_{S_1^M} u(M, t) d\omega = u(M, t) \leftarrow \boxed{\text{indep. of } \theta \text{ and } \phi}.$$

Furthermore, let V_r^M denote the ball with center M and radius r , then the volume integral on V_r^M can be expressed in spherical coordinates as:

$$\iiint_{V_r^M} f dV_r^M \stackrel{\text{Onion-peeling for integration}}{=} \int_0^r dr_1 \iint_{S_{r_1}^M} f dS_{r_1}^M = \int_0^r dr_1 \iint_{S_1^M} f(M + r_1\omega) r_1^2 d\omega.$$

Claim (Proof of (?1)). u solves $\partial_t^2 u = \Delta u \Rightarrow r\bar{u}$ solves $\partial_t^2(r\bar{u}) = \partial_r^2(r\bar{u})$. (For simplicity, take $a = 1$)

Proof. First, we integrate the wave equation on V_r^M .

$$\underbrace{\int_{V_r^M} \partial_t^2 u \, dV}_{\text{LHS}} = \underbrace{\int_{V_r^M} \Delta u \, dV}_{\text{RHS}}$$

Then proceed the RHS by **Guass formula**, while the LHS by **onion-peeling for integration**:

$$\begin{aligned} \text{RHS} &\stackrel{\text{Guass}}{=} \int_{\partial V_r^M} \mathbf{n} \cdot \nabla u \, dS = \int_{S_1^M} \partial_r u \, dS = \int_{S_1^M} \partial_r u \cdot r^2 \, d\omega = r^2 \partial_r \left(\int_{S_1^M} u(M + r\omega) \, d\omega \right) = 4\pi r^2 \partial_r \bar{u}(r, t); \\ \text{LHS} &= \partial_t^2 \int_{V_r^M} u \, dV \stackrel{\text{Onion-peeling for integration}}{=} \partial_t^2 \left(\int_0^r \int_{S_1^M} u(M + r_1\omega) r_1^2 \, d\omega \, dr_1 \right) = 4\pi \partial_t^2 \int_0^r r_1^2 \bar{u}(r_1, t) \, dr_1. \end{aligned}$$

where $\mathbf{n}$ is the outward normal. This implies

$$4\pi \partial_t^2 \int_0^r r_1^2 \bar{u}(r_1, t) \, dr_1 = 4\pi r^2 \partial_r \bar{u}(r, t)$$

Taking the derivative on both sides with respect to r

$$\Rightarrow \underbrace{\partial_t^2 \partial_r \left(\int_0^r r_1^2 \bar{u}(r_1, t) \, dr_1 \right)}_{=r^2 \partial_t^2 \bar{u}} = \underbrace{\partial_r (r^2 \partial_r \bar{u}(r, t))}_{\stackrel{\text{by (3)}}{=} r \partial_r^2 (r \bar{u})}$$

Cancel one r

$$\Rightarrow \partial_t^2 (r \bar{u}) = \partial_r^2 (r \bar{u})$$

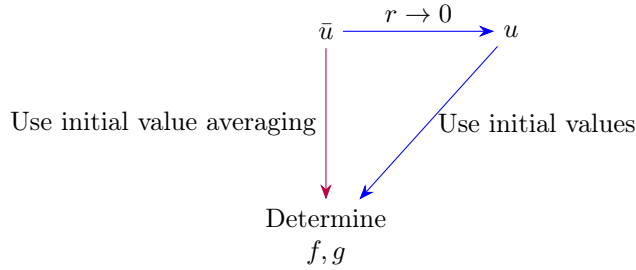
We finish the proof of this claim. □

That is, we arrive at $(r \bar{u})_{tt} = a^2 (r \bar{u})_{rr}$. Therefore, the general solution for $r \bar{u}$ is

$$r \bar{u} = f(r + at) + g(r - at), \tag{5}$$

where f and g are twice continuously differentiable functions.

There are two approaches that can utilize initial values to determine f and g according to the following flowchart.



(M1) **The blue line:** Feasible, use L'Hospital's rule to make preliminary assumptions about f and g .

(M2) **The red line:** More straightforward.

(Determine f' and g') To apply Method (M2), we need to compute the **spherical average** of the **initial values**. First, we need to satisfy the initial conditions. From (5), i.e.,

$$r \bar{u} = f(r + at) + g(r - at),$$

we obtain

$$(r \bar{u})_t = r \bar{u}_t = a f'(r + at) - a g'(r - at). \tag{6}$$

- In the form of $f + g$, $f' - g'$, to find f and g , we only need to integrate $f' - g'$, or differentiate $f + g$. We use the method of differentiation for $f + g$, otherwise it's not easy to do.
- To differentiate $f + g$, we can only differentiate $f' + g'$ with respect to r .

Let use differentiate (5) with respect to r ,

$$(r\bar{u})_r = \bar{u} + r\bar{u}_r = f'(r + at) + g'(r - at) \quad (7)$$

By (6) and (7), we calculate

$$\begin{aligned} 2f'(r + at) &= (r\bar{u})_r + \frac{1}{a}(r\bar{u})_t = \frac{\partial}{\partial r} \left(\frac{r}{4\pi} \int_{S_1^m} u \, d\omega \right) + \frac{1}{a} \frac{r}{4\pi} \int_{S_1^m} u_t \, d\omega \\ 2g'(r - at) &= (r\bar{u})_r - \frac{1}{a}(r\bar{u})_t = \frac{\partial}{\partial r} \left(\frac{r}{4\pi} \int_{S_1^m} u \, d\omega \right) - \frac{1}{a} \frac{r}{4\pi} \int_{S_1^m} u_t \, d\omega \end{aligned}$$

If $t = 0$, we obtain

$$\begin{aligned} 2f'(r) &= \frac{\partial}{\partial r} \left(\frac{r}{4\pi} \int_{S_1^m} \varphi(M + r\omega) \, d\omega \right) + \frac{1}{a} \frac{r}{4\pi} \int_{S_1^m} \psi(M + r\omega) \, d\omega \\ 2g'(r) &= \frac{\partial}{\partial r} \left(\frac{r}{4\pi} \int_{S_1^m} \varphi(M + r\omega) \, d\omega \right) - \frac{1}{a} \frac{r}{4\pi} \int_{S_1^m} \psi(M + r\omega) \, d\omega \end{aligned}$$

(Obtain u) On the other hand, using (7) (since we are considering the classical solution, $|\bar{u}_r| < \infty$ and $|\bar{u}_t| < \infty$),

$$\begin{aligned} u(M, t) &= \lim_{r \rightarrow \infty} \bar{u}(r, t) = f'(at) + g'(-at) \leftarrow \boxed{\text{or L'Hospital law by (5)}} \\ &= \frac{1}{2} \frac{\partial}{\partial t} \left(\frac{t}{4\pi} \int_{S_1^m} \varphi(M + at\omega) \, d\omega \right) + \frac{1}{2} \frac{t}{4\pi} \int_{S_1^m} \psi(M + at\omega) \, d\omega \\ &\quad + \frac{1}{2} \frac{\partial}{\partial t} \left(\frac{t}{4\pi} \int_{S_1^m} \varphi(M + at\omega) \, d\omega \right) + \frac{1}{2} \frac{t}{4\pi} \int_{S_1^m} \psi(M + at\omega) \, d\omega \\ &= \frac{\partial}{\partial t} \left(\frac{t}{4\pi} \int_{S_1^m} \varphi(M + at\omega) \, d\omega \right) + \frac{t}{4\pi} \int_{S_1^m} \psi(M + at\omega) \, d\omega \\ &= \frac{\partial}{\partial t} \left(\frac{t}{4\pi a^2 t^2} \int_{S_{at}^M} \varphi(\xi, \eta, \xi) \, dS \right) + \frac{t}{4\pi a^2 t^2} \int_{S_{at}^M} \psi(\xi, \eta, \xi) \, dS. \end{aligned} \quad (8)$$

Then

$$\begin{aligned} u(M, t) &= \underbrace{\frac{\partial}{\partial t} \left(\frac{t}{4\pi a^2 t^2} \int_{S_{at}^M} \varphi(\xi, \eta, \xi) \, dS \right)}_{\partial_t(t \times \text{the spherical average of the initial displacement on the } at \text{ sphere})} \\ &\quad + \underbrace{\frac{t}{4\pi a^2 t^2} \int_{S_{at}^M} \psi(\xi, \eta, \xi) \, dS}_{t \times \text{the spherical average of the initial velocity on the sphere of radius } at} \end{aligned}$$

Another way which is not quite direct

In (6) and (7), let $r \rightarrow 0$, we obtain

$$f'(at) = g'(-at),$$

$$u(M, t) = \lim_{r \rightarrow 0} \bar{u}(r, t) = f'(at) + g'(-at) = 2f'(at).$$

In equations (6) and (7), taking $t = 0$ gives

$$(r\bar{u})_t \big|_{t=0} = af'(r) - ag'(r), \quad (r\bar{u})_r \big|_{t=0} = f'(r) + g'(r).$$

Then we arrive at

$$\begin{aligned} 2f'(r) &= (r\bar{u})_r \big|_{t=0} + \frac{1}{a} (r\bar{u})_t \big|_{t=0} \\ &= \frac{\partial}{\partial r} \left(\frac{r}{4\pi r^2} \iint_{S_r^M} u \big|_{t=0} dS_r^M \right) + \frac{r}{a} \left(\frac{1}{4\pi r^2} \iint_{S_r^M} u_t \big|_{t=0} dS_r^M \right) \\ &= \frac{\partial}{\partial r} \left(\frac{r}{4\pi r^2} \iint_{S_r^M} \varphi(P) dS_r^M \right) + \frac{r}{a} \left(\frac{1}{4\pi r^2} \iint_{S_r^M} \psi(P) dS_r^M \right) \end{aligned}$$

Taking $r = at$ and substituting $u(M, t) = 2f'(at)$ gives

$$\begin{aligned} u(M, t) = 2f'(at) &= \frac{\partial}{\partial t} \left(\frac{t}{4\pi a^2 t^2} \iint_{S_{at}^M} \varphi(\xi, \eta, \xi) dS \right) + \frac{t}{4\pi a^2 t^2} \iint_{S_{at}^M} \psi(\xi, \eta, \xi) dS \\ &= \frac{\partial}{\partial t} \left(\frac{t}{4\pi} \iint_{S_1^M} \varphi(M + at\omega) d\omega \right) + \frac{t}{4\pi} \iint_{S_1^M} \psi(M + at\omega) d\omega. \end{aligned}$$

When the initial functions are sufficiently smooth, it is easy to verify that the function $u(x, y, z, t)$ represented by formula (8) is indeed the solution to problem (1)–(2).

Ex 0.1. Solve the following initial value problem

$$\begin{cases} u_{tt} = u_{xx} + u_{yy} + u_{zz} & (-\infty < x, y, z < +\infty, t > 0), \\ u(x, y, z, 0) = 0, & u_t(x, y, z, 0) = 2xy, \end{cases}$$

Solution. From formula (8), we get

$$\begin{aligned} u(\underbrace{x, y, z, t}_{\text{Your position}}) &= \frac{t}{2\pi} \int_0^{2\pi} \int_0^\pi \underbrace{(x + t \sin \theta \cos \varphi)(y + t \sin \theta \sin \varphi) \sin \theta}_{\text{Sphere around you}} d\theta d\varphi \\ &= \frac{t}{2\pi} \int_0^{2\pi} \int_0^\pi (xy \sin \theta + xt \sin^2 \theta \sin \varphi + yt \sin^2 \theta \cos \varphi + t^2 \sin^3 \theta \cos \varphi \sin \varphi) d\theta d\varphi \\ &= \frac{t}{2\pi} \int_0^{2\pi} \int_0^\pi (x + t \sin \theta \cos \varphi)(y + t \sin \theta \sin \varphi) \sin \theta d\theta d\varphi \\ &= \frac{xyt}{2\pi} \int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\varphi = 2xyt. \end{aligned}$$

3.2.2 Dimension Reduction Method

Using the dimension reduction method to solve the initial value problem of the two-dimensional wave equation.

$$\begin{cases} u_{tt} = a^2(u_{xx} + u_{yy}) & (-\infty < x, y < +\infty, t > 0), \\ u|_{t=0} = \varphi(x, y), \\ u_t|_{t=0} = \psi(x, y). \end{cases} \quad (9)$$

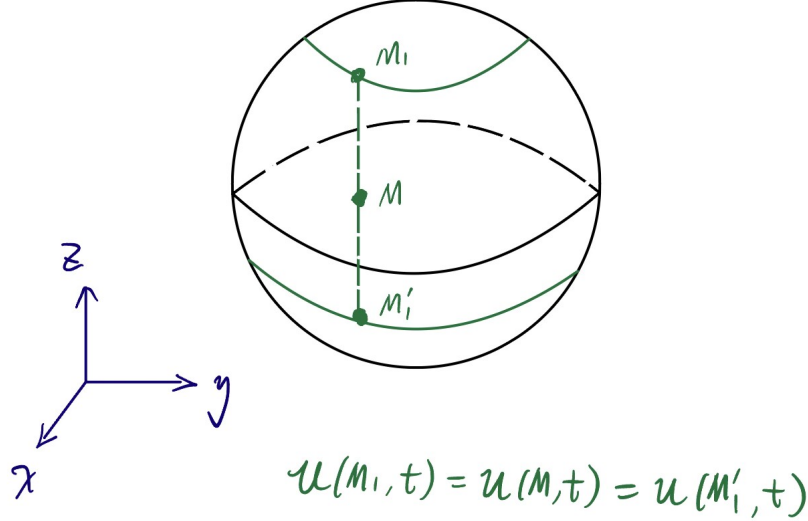


Figure 3: 2D wave equations

Since the initial value problem of the two-dimensional wave equation can be considered a special case of the initial value problem of the three-dimensional wave equation (see Fig. 3), the Poisson formula for the three-dimensional wave equation can be used to represent the solution of the initial value problem of the two-dimensional wave equation, and thus derive another form of the solution representation for the two-dimensional problem.

- The 2D problem is represented on a plane. Imagine a large trampoline with no boundaries, vibrating in two dimensions.
- To convert this 2D problem into a 3D case, we **extend the 2D surface along the z -axis**.
- This extension involves replicating the 2D surface infinitely along the z -axis, keeping the value of u constant across different z levels.
- After the surface is stacked infinitely, it forms a 3D structure, and we can apply 3D formulas such as Kirchhoff's equations.
- The **key idea** is that along the z -axis, $u_{zz} = 0$ because the value of u does not change in the z -direction.
- This extension allows us to use the 3D Kirchhoff equation to solve the problem.
- To calculate any point in the stacked structure, we apply the second-order wave equation, averaging over a spherical surface surrounding the point.
- Finally, we need to project the result back onto the original 2D trampoline.

Mathematical Interpretation

- The value of u is constant along the z -axis, leading to $u_{zz} = 0$.

Claim.

Initial data is independent of z , implies u is independent of z for all $t > 0$.

Proof. Given $u_z|_{t=0} = 0$, $\partial_t u_z|_{t=0} = 0$, differentiating the wave equation with respect to z ,

$$(u_z)_{tt} = a^2((u_z)_{xx} + (u_z)_{yy}) \Rightarrow u_z \equiv 0.$$

□

- Once we reduce the problem to 3D, we can apply the 3D Kirchhoff equation directly to solve for $u(t)$.

Using Equation (8)

$$u(M, t) = \frac{\partial}{\partial t} \left(\frac{t}{4\pi a^2 t^2} \iint_{S_{at}^M} \varphi(\xi, \eta, \zeta) dS \right) + \frac{t}{4\pi a^2 t^2} \iint_{S_{at}^M} \psi(\xi, \eta, \zeta) dS$$

The solution to the initial value problem of the two-dimensional wave equation (9) can be obtained as

$$u(x, y, t) = \frac{\partial}{\partial t} \left(\frac{t}{4\pi a^2 t^2} \iint_{S_{at}^M} \varphi dS \right) + \frac{t}{4\pi a^2 t^2} \iint_{S_{at}^M} \psi dS,$$

where the integration is performed on the sphere S_{at}^M in the three-dimensional space (x, y, z) .

- After extending the 2D surface to 3D, the task is to project the 3D spherical surface integral back onto the 2D trampoline surface.
- This involves using a projection formula for surface integrals that you learned in calculus.
- The formula for projecting a surface integral is:

$$\int_{\Sigma} f d\sigma = \int_S f \cos \gamma dS$$

where γ is the angle between the surface normals.

- This projection simplifies the surface integral from 3D to 2D.
- Alternatively, using the **standard method** of **calculating surface integrals** in calculus. If $z = \varphi(x, y)$, the integral can be transformed to:

$$\int_S f(x, y) dS = \int_{\Sigma} f(x, y) \sqrt{1 + \left(\frac{\partial \varphi}{\partial x} \right)^2 + \left(\frac{\partial \varphi}{\partial y} \right)^2} dx dy$$

where $f(x, y)$ represents the integrand function and the square root term adjusts for the curvature of the surface. For a sphere, $z = \varphi(x, y) = \sqrt{(at)^2 - (\xi - x)^2 - (\eta - y)^2}$, then

$$(\varphi_x)^2 = \frac{(\xi - x)^2}{(at)^2 - (\xi - x)^2 - (\eta - y)^2} \quad \text{and} \quad (\varphi_y)^2 = \frac{(\eta - y)^2}{(at)^2 - (\xi - x)^2 - (\eta - y)^2}$$

Then (see Fig. 3)

$$dS = \sqrt{1 + \left(\frac{\partial\varphi}{\partial x}\right)^2 + \left(\frac{\partial\varphi}{\partial y}\right)^2} d\sigma = \frac{at}{\sqrt{(at)^2 - (\xi - x)^2 - (\eta - y)^2}} d\sigma$$

- This formula is derived from standard techniques in multivariable calculus and is applicable for transforming integrals from a curved surface to a flat 2D plane.

Key Concepts

- Projection of 3D integrals onto 2D involves transforming the integrand using standard surface integration formulas.
- The formula for surface integrals in calculus allows us to compute the projected integral over a 2D region from a 3D surface.
- A key aspect is recognizing how surface curvature and the normal vector impact the projection of the integral.

Since φ and ψ are functions independent of z , the integral on the spherical surface can be transformed into the integral on its projection on the plane $z = \text{constant}$: $\Sigma_{at}^M : (\xi - x)^2 + (\eta - y)^2 \leq a^2 t^2$. Since the area element dS on the spherical surface and its projection area element $d\sigma$ satisfy the following relationship (see Fig. 3):

$$d\sigma = \cos \gamma \cdot dS,$$

where γ is the angle between the normal directions of these two surface elements. Therefore, we have:

$$\cos \gamma = \frac{\sqrt{(at)^2 - (\xi - x)^2 - (\eta - y)^2}}{at}.$$

Note that the integrals over the upper and lower hemispheres both transform into integrals over the same circle, hence the integral over the circle Σ_{at}^M should be **taken as twice the integral over the upper hemisphere**.

Thus,

$$\begin{aligned} u(x, y, t) &= \frac{\partial}{\partial t} \left(\frac{t}{4\pi a^2 t^2} \iint_{\Sigma_{at}^M} \varphi dS \right) + \frac{t}{4\pi a^2 t^2} \iint_{\Sigma_{at}^M} \psi dS \\ &= \frac{\partial}{\partial t} \left(\frac{t}{4\pi a^2 t^2} \iint_{\Sigma_{at}^M} \frac{2\varphi}{\cos \gamma} d\sigma \right) + \frac{t}{4\pi a^2 t^2} \iint_{\Sigma_{at}^M} \frac{2\psi}{\cos \gamma} d\sigma, \\ u(x, y, t) &= \frac{1}{2\pi a} \frac{\partial}{\partial t} \left[\iint_{\Sigma_{at}^M} \frac{\varphi(\xi, \eta) d\sigma}{\sqrt{(at)^2 - (\xi - x)^2 - (\eta - y)^2}} \right] + \frac{1}{2\pi a} \iint_{\Sigma_{at}^M} \frac{\psi(\xi, \eta) d\sigma}{\sqrt{(at)^2 - (\xi - x)^2 - (\eta - y)^2}}. \end{aligned} \quad (10)$$

The above equation is called the **Poisson formula** for the initial value problem of the two-dimensional wave equation. Since the integration region $\Sigma_{at}^M : (\xi - x)^2 + (\eta - y)^2 \leq a^2 t^2$ is a circular domain centered at M with radius at , we usually use **polar coordinates** to calculate the integral in equation (10).

Ex 0.2. Solve the following problem

$$\begin{cases} u_{tt} = u_{xx} + u_{yy} & (-\infty < x, y < +\infty, t > 0), \\ u|_{t=0} = 0, \quad u_t|_{t=0} = 2xy. \end{cases}$$

Solution. From equation (10), we get

$$\begin{aligned} u(x, y, t) &= \frac{1}{2\pi} \int_0^t \int_0^{2\pi} \frac{2(x + \rho \cos \theta)(y + \rho \sin \theta)}{\sqrt{t^2 - \rho^2}} \rho d\rho d\theta \\ &= \frac{xy}{\pi} \int_0^t \int_0^{2\pi} \frac{\rho d\rho d\theta}{\sqrt{t^2 - \rho^2}} = 2xyt. \end{aligned}$$

Key Differences Between 2D and 3D Solution Formulas

- In 3D, the integral is taken over a **hollow spherical surface**, meaning the integration surface is a sphere with an empty interior.
- In 2D, the integral is taken over a **solid circular region**, meaning the entire disk, including its interior, contributes to the integration.
- This **difference** is a fundamental characteristic to remember when working with wave equations in different dimensions.

Modification of the Integrand

- In 3D, the integrand function is simply φ, ψ .
- In 2D, the integrand includes an additional factor due to the transformation of coordinates.
- As time increases, the denominator of the modified integrand grows, leading to a decrease in the integral's value.
- This results in **attenuation over time** in 2D wave propagation.

3.2.3 Physical Significance of the Solution

Physical Interpretation

- Consider a region Ω where an initial perturbation is introduced.
- Outside Ω , the initial state is zero.
- Observing from a point M , one will detect the perturbation after some time as the wave propagates.
- Wave propagation speed is finite, meaning there is a delay before the perturbation reaches M .
- At time T_1 , the value at M is determined by an **average over a spherical shell (3D) or a circular disk (2D)**.
- The integration method used depends on whether the problem is in 2D or 3D.

Recall Solution (8)

$$u(M, t) = \frac{\partial}{\partial t} \left(\frac{t}{4\pi a^2 t^2} \iint_{S_{at}^M} \varphi(\xi, \eta, \zeta) dS \right) + \frac{t}{4\pi a^2 t^2} \iint_{S_{at}^M} \psi(\xi, \eta, \zeta) dS.$$

Assume that the initial disturbance occurs only within a finite region Ω in space. Outside of region Ω , consider any point M , and examine the situation of the influence of the initial disturbance on point M at different times (see Fig. 4).

We know that the value of the solution u at point M and time t , $u(M, t)$, is determined by the values of the initial functions φ and ψ on the sphere S_{at}^M . Therefore, the integral in equation (8) is not zero **only** when the sphere S_{at}^M **intersects** with region Ω , hence $u(M, t) \neq 0$.

Let d and D represent the **closest** and **farthest** distances from point M to region Ω , respectively, as shown in the figure. When $at < d$, the sphere S_{at}^M is still at a distance from region Ω , so the values of φ and ψ on this sphere are 0, the integral is 0, and thus $u(M, t) = 0$. At this time, the disturbance has not yet reached point M (see Fig. 5).

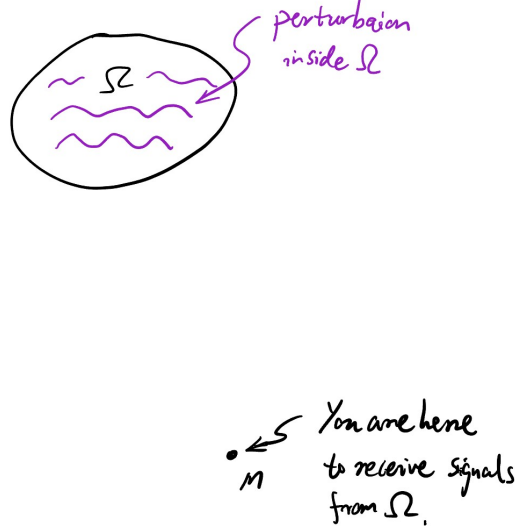


Figure 4: Physical meaning 1

When $d \leq at \leq D$, the sphere S_{at}^M continuously intersects with region Ω , the value of the integral is generally not 0, and the value of $u(M, t)$ is generally not 0 either. At this time, point M is in a disturbed state. The initial disturbance reaches point M instantaneously at $t = d/a$ (see Fig. 6)..

When $at > D$, the sphere S_{at}^M has already passed the initial disturbance region Ω and no longer intersects with it. Starting from $t = D/a$, $u(M, t)$ takes the zero value again, indicating that the disturbance has already passed point M , and point M returns to its original state of rest (see Fig. 7).

In a bounded region Ω , any disturbance caused by a point propagates outward at speed a . Therefore, at time t , the **region affected by the initial disturbance** in Ω is **all of the spheres** centered at $p \in \Omega$ with radius at . When t is sufficiently large, these spherical surfaces have **two envelopes**. The **outer envelope** is called the **front wavefront**, and the **inner envelope** is called the **rear wavefront**. The middle part between these two wavefronts is the region affected by the initial disturbance.

From the above analysis, it can be seen that the wave generated by a point disturbance propagates as a spherical surface passing through M . A large number of point disturbances form the entirety of these spherical surfaces, which together constitute the disturbance region. Between the disturbance regions, there exist a leading envelope and a trailing envelope.

The part outside the front wavefront indicates the region where the wave has **not yet reached**, while the part inside the rear wavefront indicates the region where the wave has **passed and returned to its original state**. Therefore, when the initial disturbance is confined to a certain local area in space, the wave propagation has a clear front and rear wavefront. This phenomenon is known in physics as **Huygens' principle** or the **non-retroactive phenomenon**. Since the disturbance at point $M_0 \in \Omega$ at $t = t_0$ affects the sphere $S_{at_0}^{M_0}$ centered at M_0 with radius at_0 , solution (8) is referred to as a **spherical wave**.

Recall Solution (10):

$$u(x, y, t) = \frac{1}{2\pi a} \frac{\partial}{\partial t} \left[\iint_{\Sigma_{at}^M} \frac{\varphi(\xi, \eta) d\sigma}{\sqrt{(at)^2 - (\xi - x)^2 - (\eta - y)^2}} \right] + \frac{1}{2\pi a} \iint_{\Sigma_{at}^M} \frac{\psi(\xi, \eta) d\sigma}{\sqrt{(at)^2 - (\xi - x)^2 - (\eta - y)^2}}.$$

For the solution of the initial value problem of the two-dimensional wave equation (10), a similar discussion can be made. However, it is important to note that since the **integration is performed over the circular domain** $\Sigma_{at}^M : (\xi - x)^2 + (\eta - y)^2 \leq a^2 t^2$. Thus for any point M , once $u(M, t)$ changes from 0 to non-zero as time t increases, it will **not gradually decrease back to 0** as in the spatial case, but it will **gradually decrease** from a certain moment onward. Thus, there are **significant differences** between two-dimensional and three-dimensional situations (see Fig. 8).

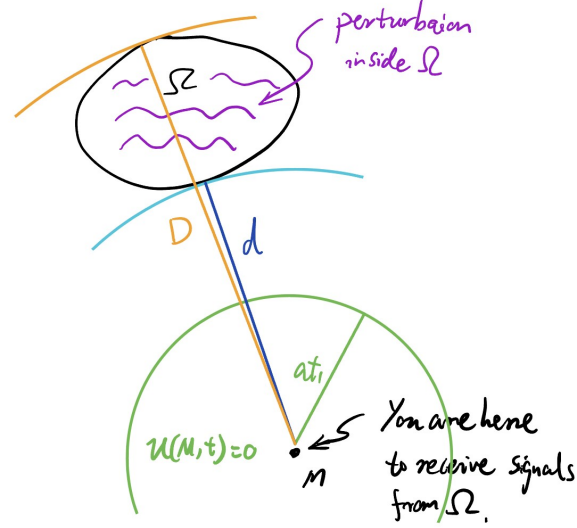


Figure 5: Physical meaning 2

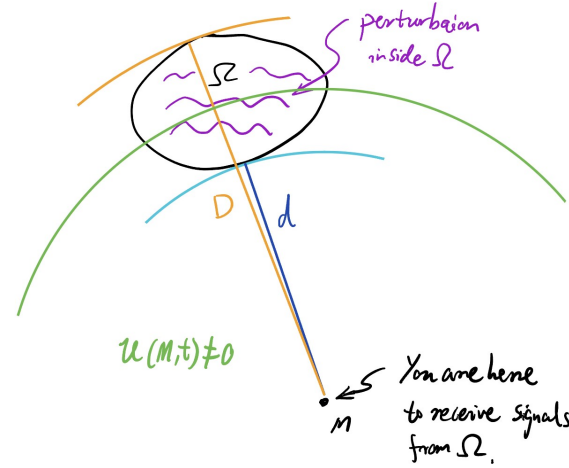


Figure 6: Physical meaning 3

For the two-dimensional case, the wave propagation **only has a front wavefront** and **no rear wavefront**, and **Huygens' principle no longer holds**. This phenomenon is called **wave diffusion**, or in other words, this type of wave exhibits a **retroactive effect**. For the two-dimensional problem, it can be considered that the initial disturbance occurs within an **infinitely long cylinder** and is **independent** of the z -coordinate. Thus, the initial disturbance at point M_0 should be considered as an initial disturbance along an infinitely long straight line passing through point M_0 and parallel to the z -axis. At $t = t_0$, its influence is within a cylindrical surface with this line as the axis and at_0 as the radius. Therefore, solution (10) is referred to as a **cylindrical wave**.

Summary

Wave Propagation and Initial Disturbance (see Fig. 4)

- Consider a region Ω with an initial disturbance.
- Outside Ω , the initial condition is zero.
- Observation point M is used to analyze wave propagation.

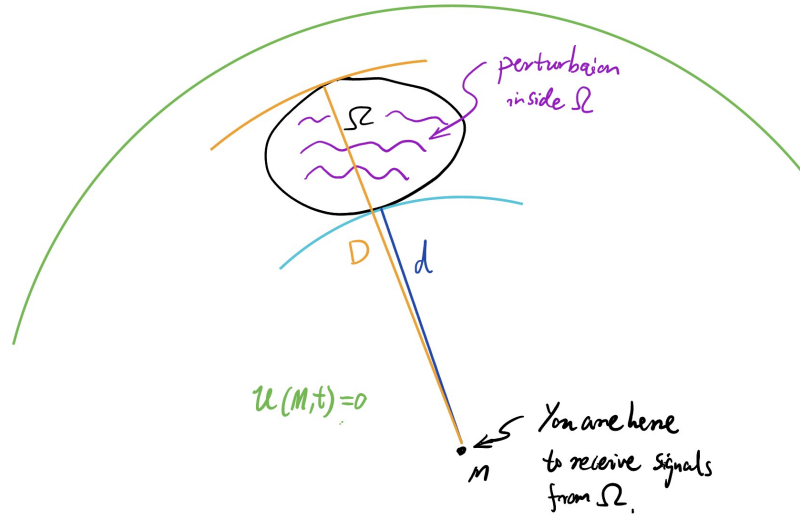


Figure 7: Physical meaning 4

- As time evolves, point M will receive the wave signal after a certain delay due to finite propagation speed.

Three Stages of Wave Reception

First Stage: No Signal (see Fig. 5)

- At time t_1 , construct a spherical surface centered at M with radius at_1 .
- If this sphere does not intersect Ω , the integral over the surface is zero.
- Thus, $u(M, t_1) = 0$, meaning no signal is received at M .

Second Stage: Signal Reception (see Fig. 6)

- At time t_2 , the spherical surface intersects Ω .
- Since the integral now includes nonzero contributions from Ω , $u(M, t_2) \neq 0$.
- This indicates that the signal has reached M .

Third Stage: Key Differences Between 2D and 3D

For 3D Waves (see Fig. 7):

- At time t_3 , the sphere extends beyond Ω .
- The integral is again zero, implying that $u(M, t_3) = 0$.
- This demonstrates the principle of **Huygens' Principle** (no after-effect phenomenon).

For 2D Waves (see Fig. 8):

- In 2D, the integration region is a solid disk instead of a spherical shell.
- At time t_3 , the integral remains nonzero since the solid region still overlaps with Ω .
- However, the signal weakens over time due to a decay factor in the denominator.
- This results in a **persistence effect** and **wave dispersion**, unlike the 3D case.

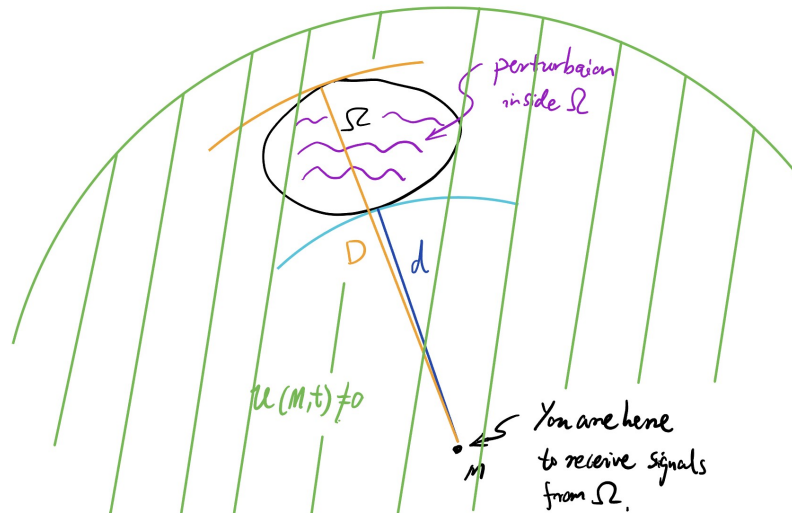


Figure 8: Physical meaning 5

Key Differences Between 2D and 3D Wave Propagation

- **3D waves** exhibit **no after-effect**: once the signal passes, the region remains undisturbed ("Let bygones be bygones").
- **2D waves** exhibit **after-effect**: signals persist beyond initial interaction but decay over time ("What's done is done").
- The difference arises from the geometry of integration: spherical surface vs. solid disk.